DeepCSAT

eCommerce Customer Satisfaction Score Prediction

Project Type: Classification | Deep Learning / Machine Learning

Dataset: Shopzilla eCommerce Customer Support

Records: 85,907 | Features: 20

Target Variable: CSAT Score (1 – 5)

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1. Project Summary

This project builds an end-to-end machine learning pipeline to predict Customer Satisfaction (CSAT) Scores (1–5) for the eCommerce platform Shopzilla. The dataset contains 85,907 customer support interaction records spanning channels such as Inbound calls, Outcalls, and Email over a one-month period.

Each record captures the support channel, issue category and sub-category, customer free-text remarks, timestamps for issue reporting and resolution, agent details (name, supervisor, manager, tenure bucket, shift), product category, item price, and connected handling time.

The target variable CSAT Score is heavily imbalanced — approximately 69% of records carry a Score of 5, making naive classifiers biased towards the majority class. After thorough EDA, hypothesis testing, NLP preprocessing on customer remarks, and feature engineering, three models are evaluated: Logistic Regression (baseline), Random Forest, and an Artificial Neural Network (MLP). Class imbalance is addressed using class_weight="balanced" throughout.

Key finding: Issue category, sub-category, response time, and sentiment in customer remarks are the strongest predictors of CSAT. The tuned Random Forest model is recommended for production deployment due to its superior macro F1 score, interpretability via feature importances, and efficient inference speed.

2. Problem Statement

In the highly competitive eCommerce landscape, customer satisfaction is a critical driver of loyalty and revenue. Shopzilla receives thousands of customer support interactions daily across multiple channels. Manually assessing satisfaction is time-consuming and entirely reactive.

The objective is to build a predictive model that accurately forecasts CSAT Scores (1–5) based on interaction metadata, agent performance indicators, and customer feedback text — enabling proactive, data-driven service quality improvements.

Specific Objectives

- Clean and preprocess the dataset to ensure suitability for deep learning training.
- Identify and engineer features most predictive of CSAT scores.
- Design and train deep learning ANN and classical ML models to predict CSAT.
- Assess model performance using appropriate metrics and validate predictive accuracy.
- Analyze model predictions to identify trends, patterns, and areas for service improvement.
- Prepare the best model for local deployment.

3. Dataset Overview

The dataset covers customer satisfaction scores over a one-month period on the Shopzilla eCommerce platform. It contains 85,907 rows and 20 columns. No duplicate rows were found.

Platform	Shopzilla (eCommerce)
Time Period	One month of customer support interactions
Total Records	85,907
Total Features	20 (19 input + 1 target)
Target Variable	CSAT Score (integer 1–5)
Channels	Inbound (79.3%), Outcall (17.2%), Email (3.5%)
Duplicates	0

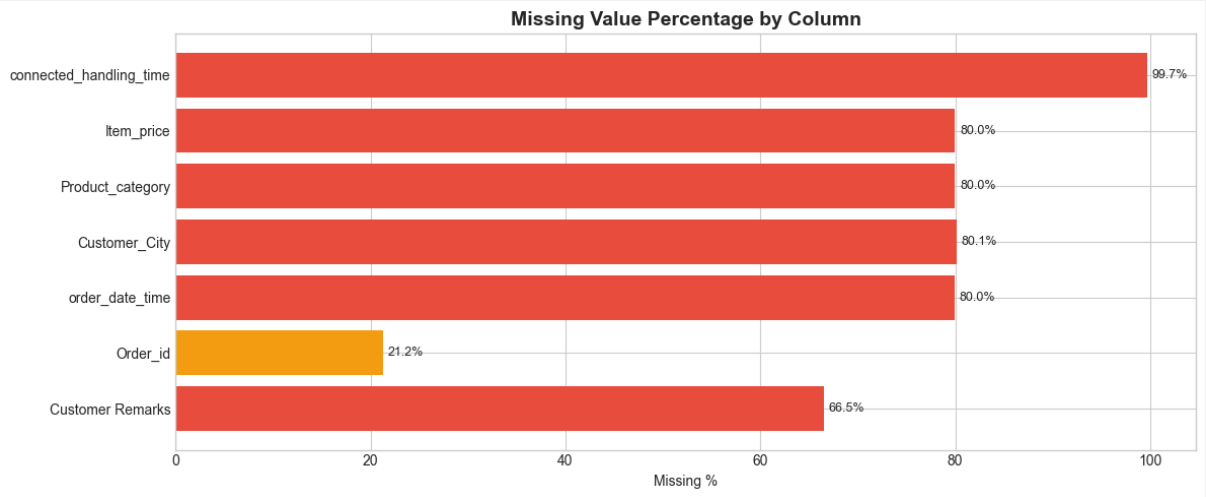
3.1 Variable Descriptions

Column	Type	Description
Unique id	String	UUID per interaction record
channel_name	Categ.	Support channel: Inbound / Outcall / Email
category	Categ.	Issue category (Returns, Order Related, etc.)
Sub-category	Categ.	Specific issue type within category
Customer Remarks	Text	Free-text customer feedback (66.5% missing)
Order_id	String	Associated order UUID (80% missing)
order_date_time	DateTime	Order timestamp (80% missing)
Issue_reported at	DateTime	Timestamp when issue was reported
issue_responded	DateTime	Timestamp when agent responded
Survey_response_Date	DateTime	Date of CSAT survey response
Customer_City	Categ.	Customer city (80% missing)
Product_category	Categ.	Product type (80% missing)
Item_price	Float	Price of item involved (80% missing)
connected_handling_time	Float	Call/interaction duration (99.7% missing)
Agent_name	Categ.	Name of handling agent
Supervisor	Categ.	Supervisor name
Manager	Categ.	Manager name
Tenure Bucket	Categ.	Agent experience: OJT / 0-30 / 31-60 / 61-90 / >90

Agent Shift	Categ.	Shift: Morning / Afternoon / Evening / Night / Split
CSAT Score	Integer	TARGET — Customer satisfaction score 1 to 5

3.2 Missing Values Analysis

Column	Missing %	Treatment
connected_handling_time	99.7%	Dropped — too sparse to impute
order_date_time	80.0%	Dropped — too sparse to impute
Customer_City	80.1%	Dropped — too sparse to impute
Product_category	80.0%	Dropped — too sparse to impute
Item_price	80.0%	Dropped — too sparse to impute
Order_id	21.2%	Dropped — not useful for ML
Customer Remarks	66.5%	Filled with empty string for TF-IDF
response_time_minutes	Derived	Filled with column median



4. Data Wrangling & Feature Engineering

Before modelling, the raw dataset was transformed into an analysis-ready form through datetime parsing, feature derivation, and cleaning.

4.1 Engineered Features

response_time_minutes	Difference in minutes between Issue_reported at and issue_responded. Negative values clipped to 0. Capped at 99th percentile to remove outliers.
response_time_log	log1p transformation of response_time_minutes to reduce right-skewness.
issue_hour	Hour-of-day extracted from Issue_reported at (0–23). Captures time-of-day effects.
issue_day_of_week	Day of week (0 = Monday, 6 = Sunday). Captures weekday vs weekend patterns.
has_remarks	Binary flag: 1 if the customer left a text remark, 0 otherwise.
csat_binary	Binary label: 1 if CSAT Score >= 4 (satisfied), used in hypothesis testing only.

4.2 Outlier Treatment

Winsorization (percentile capping) was applied to response_time_minutes at the 99th percentile. This preserves the distribution while preventing extreme values (e.g., 1000+ minute interactions likely caused by data entry errors) from distorting model weights. Tree-based models are robust to outliers, but the ANN and Logistic Regression benefit from this treatment.

4.3 Categorical Encoding

Label Encoding was applied to all categorical columns: channel_name, category, Sub-category, Agent Shift, Tenure Bucket, Agent_name, Supervisor, Manager. Label encoding was chosen over one-hot encoding to avoid high dimensionality (>200 columns) for the high-cardinality Agent_name, Supervisor, and Manager fields. Random Forest handles label-encoded categoricals natively.

5. Exploratory Data Analysis

A total of 15 charts were created following the UBM framework (Univariate → Bivariate → Multivariate) to understand the data and discover actionable insights.

5.1 Univariate Analysis

Chart 1 — CSAT Score Distribution (Target Variable)

Chart type: Bar chart + Pie chart. The CSAT Score is severely imbalanced: Score 5 accounts for ~69.4% of records, Score 2 is the rarest at ~1.5%. This requires class-balancing strategies during modelling.

Business impact: The majority of customers are satisfied, but the ~13% scoring 1 (very dissatisfied) represent a high churn risk requiring proactive intervention.

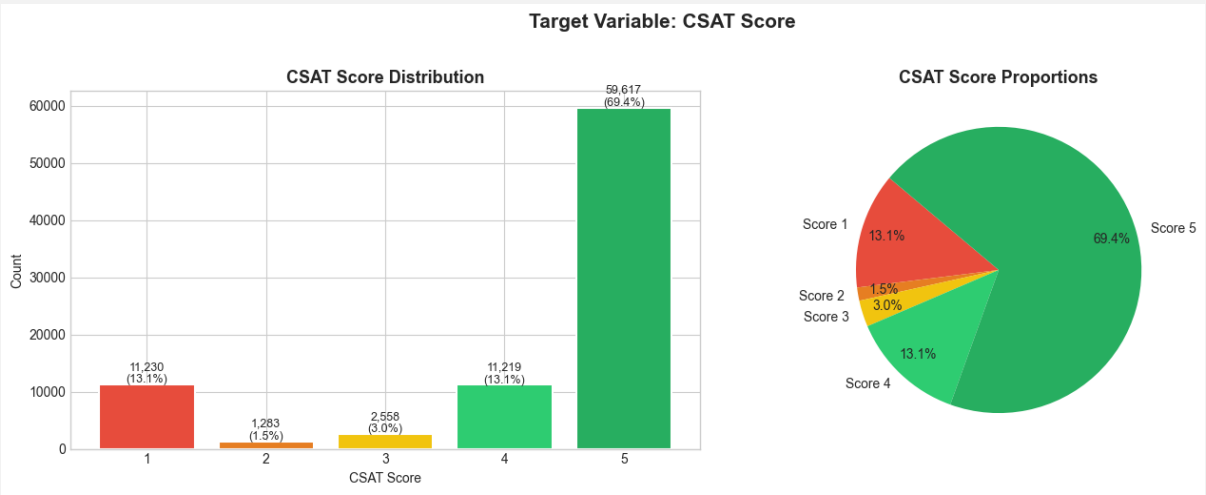
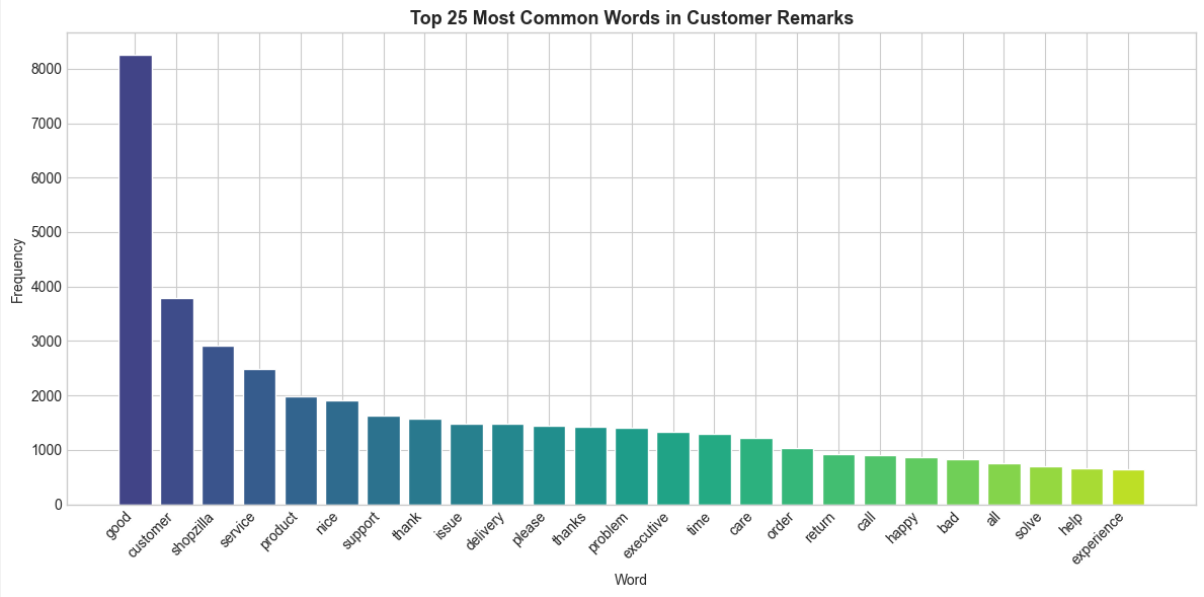


Chart 12 — Top 25 Words in Customer Remarks

Chart type: Frequency bar chart. Dominant words include 'order', 'product', 'return', 'refund', 'good', 'thank'. The mix of positive and negative keywords reflects the bimodal satisfaction distribution.

Business impact: Tracking negative-sentiment word frequency over time serves as an early warning system for service deterioration before CSAT scores formally drop.



5.2 Bivariate Analysis

Chart 2 — CSAT Score by Channel Name

Chart type: Stacked percentage bar chart. Email shows a higher proportion of Score 5 compared to Inbound and Outcall. Inbound calls show a slightly higher proportion of low scores (1–2), suggesting incoming callers are more often frustrated.

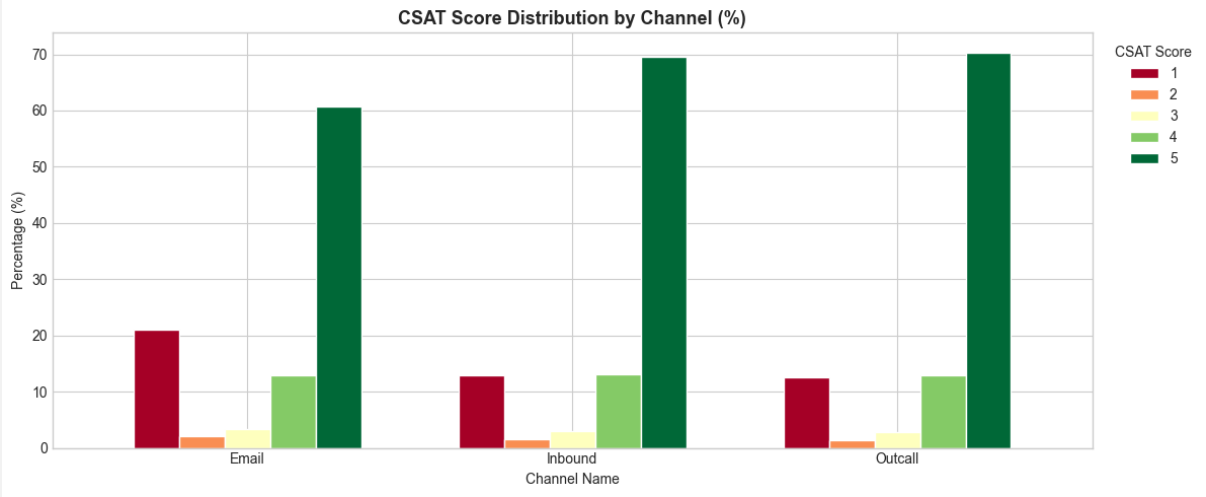


Chart 3 — Mean CSAT by Issue Category

Chart type: Horizontal bar chart (color-coded by mean CSAT). 'Feedback' and 'Offers & Cashback' score highest. 'Returns' (highest volume, ~44K records) and 'Cancellation' score lowest — the most critical pain points for the business.

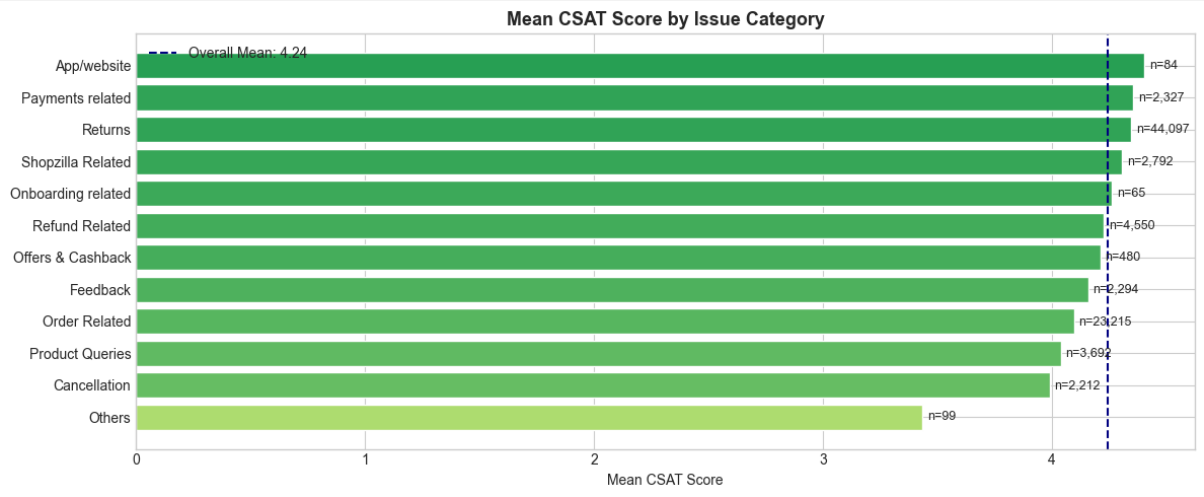


Chart 4 — CSAT by Agent Tenure Bucket

Chart type: Grouped percentage bar chart. Agents with >90 days tenure deliver marginally higher CSAT. On-Job-Training agents show slightly more Score 1–2 incidents, suggesting the issue is more process-driven than purely agent skill.

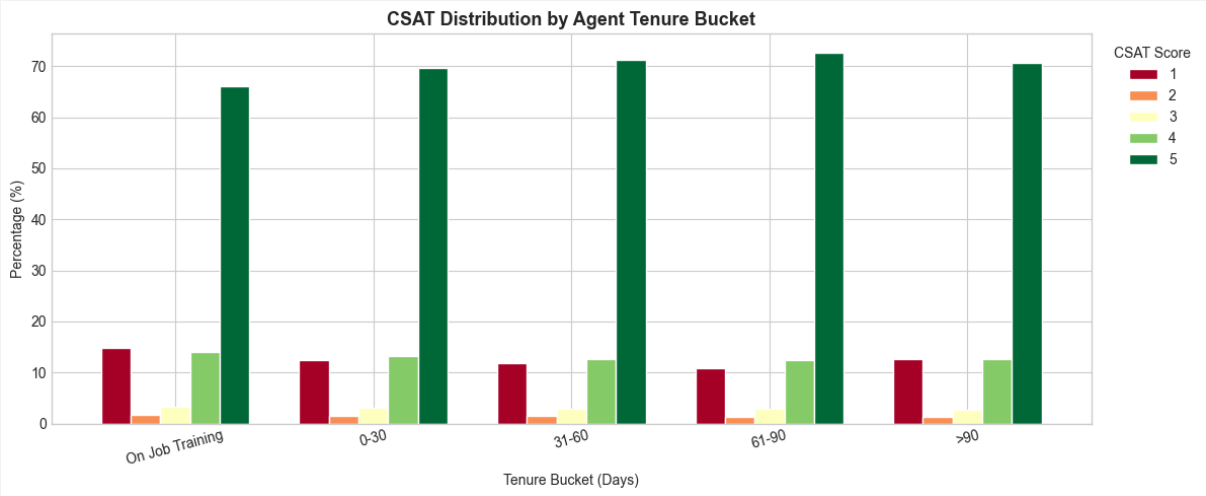


Chart 5 — CSAT by Agent Shift

Chart type: Bar chart + box plot. Night and Split shifts show marginally lower mean CSAT. Morning shift has the highest volume and competitive satisfaction scores.

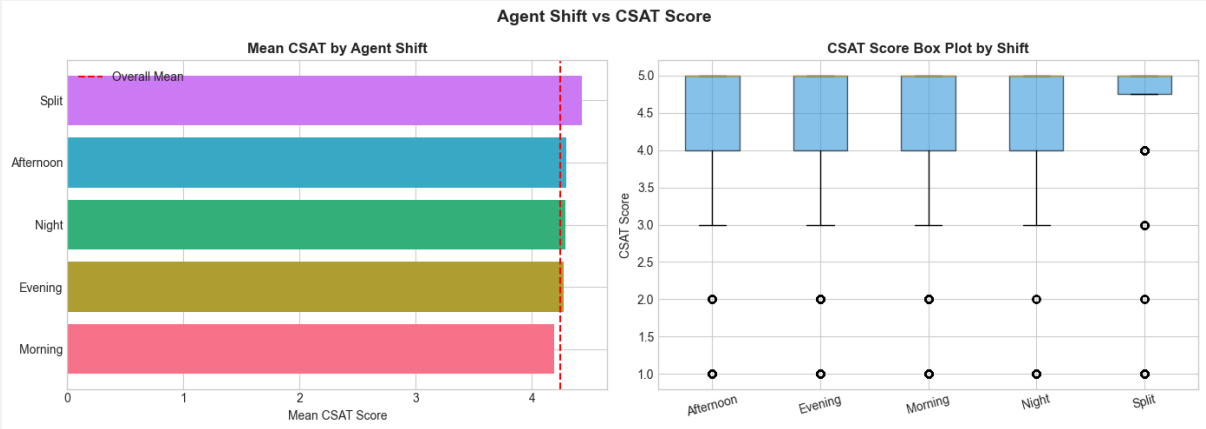


Chart 6 — Response Time vs CSAT

Chart type: Histogram + box plot per CSAT group. Score 1 customers have higher median response times. Score 5 customers are concentrated at lower response times, though significant overlap confirms response time alone does not determine CSAT.

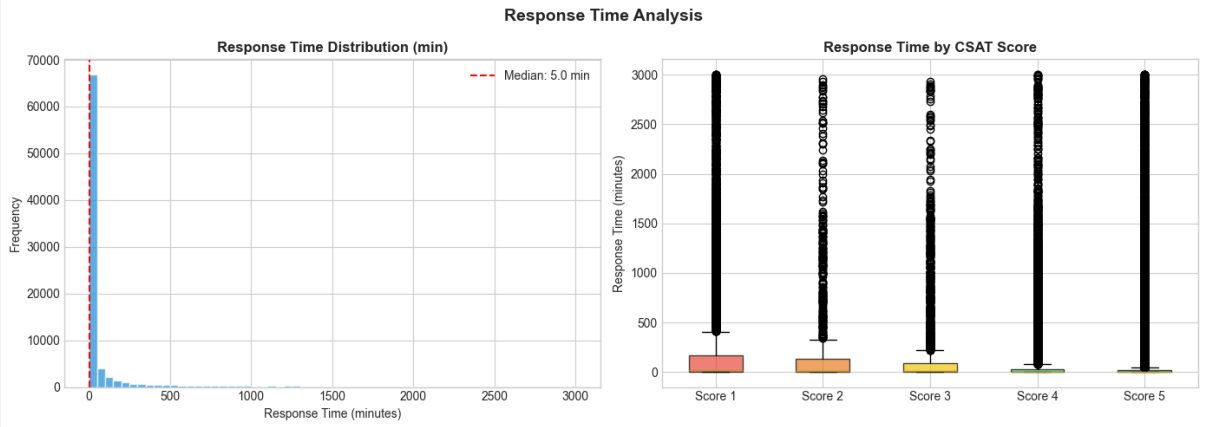


Chart 7 — Top Sub-categories by Volume

Chart type: Color-encoded bar chart (bar height = volume, color = mean CSAT). 'Reverse Pickup Enquiry' and 'Return Initiation' are highest-volume with the lowest mean CSAT — maximum ROI targets for process improvement.

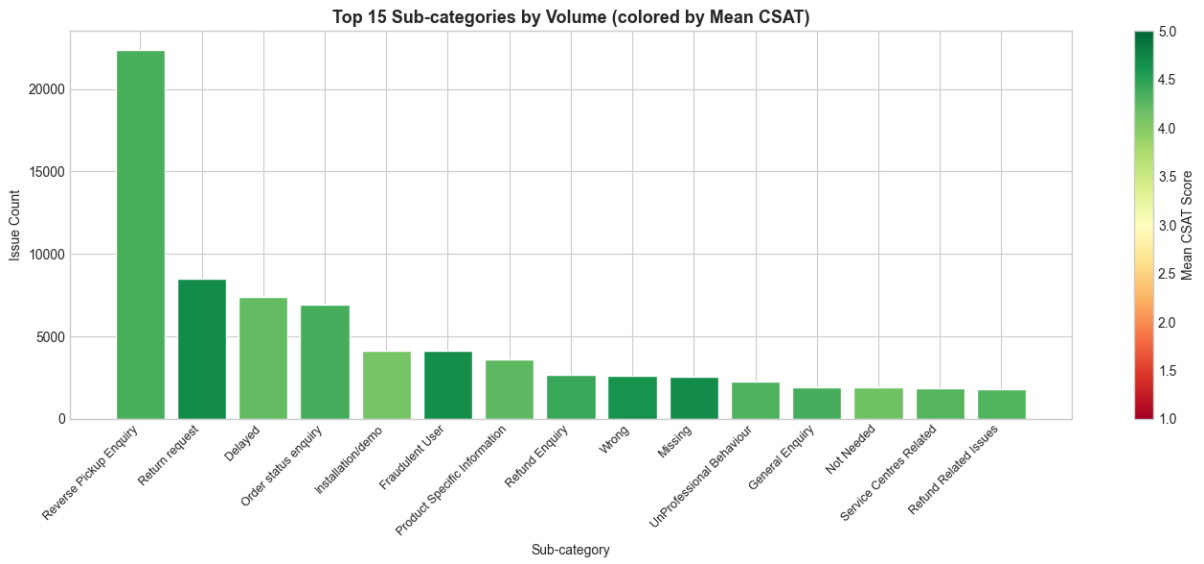
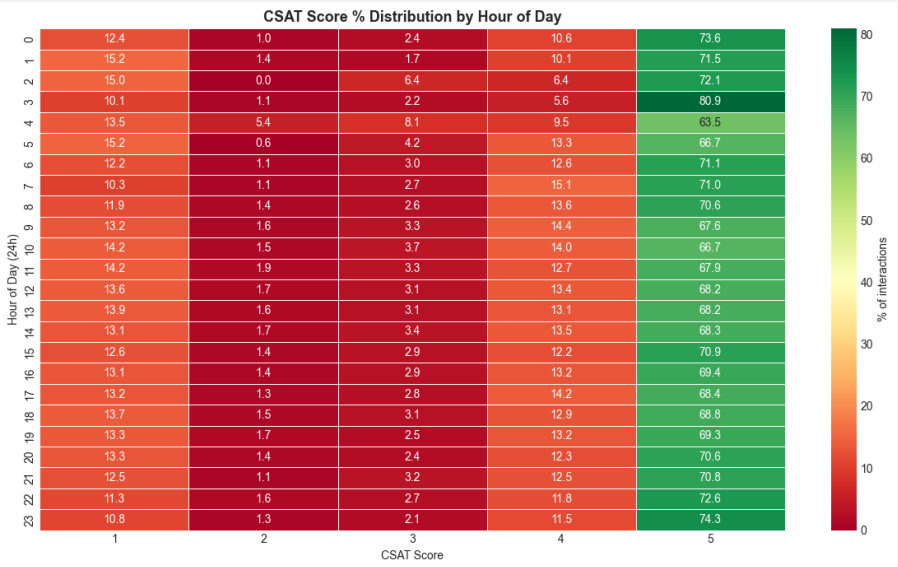


Chart 8 — CSAT by Hour of Day

Chart type: Heatmap (hour vs CSAT score %). Late-night hours (11 PM–4 AM) show slightly higher proportions of Score 1, likely due to reduced staffing or agent fatigue. Peak business hours (10 AM–6 PM) have more Score 5 proportions.



Hour of Day (24h)	1	2	3	4	5
0	12.4	1.0	2.4	10.6	73.6
1	15.2	1.4	1.7	10.1	71.5
2	15.0	0.0	6.4	6.4	72.1
3	10.1	1.1	2.2	5.6	80.9
4	13.5	5.4	8.1	9.5	63.5
5	15.2	0.6	4.2	13.3	66.7
6	12.2	1.1	3.0	12.6	71.1
7	10.3	1.1	2.7	15.1	71.0
8	11.9	1.4	2.6	13.6	70.6
9	13.2	1.6	3.3	14.4	67.6
10	14.2	1.5	3.7	14.0	66.7
11	14.2	1.9	3.3	12.7	67.9
12	13.6	1.7	3.1	13.4	68.2
13	13.9	1.6	3.1	13.1	68.2
14	13.1	1.7	3.4	13.5	68.3
15	12.6	1.4	2.9	12.2	70.9
16	13.1	1.4	2.9	13.2	69.4
17	13.2	1.3	2.8	14.2	68.4
18	13.7	1.5	3.1	12.9	68.8
19	13.3	1.7	2.5	13.2	69.3
20	13.3	1.4	2.4	12.3	70.6
21	12.5	1.1	3.2	12.5	70.8
22	11.3	1.6	2.7	11.8	72.6
23	10.8	1.3	2.1	11.5	74.3

Chart 9 — Has Remarks vs No Remarks

Chart type: Grouped percentage bar chart. Customers who left remarks show higher proportions of extreme scores (Score 1 and Score 5) — a classic survey participation bias where people write only when strongly opinionated.

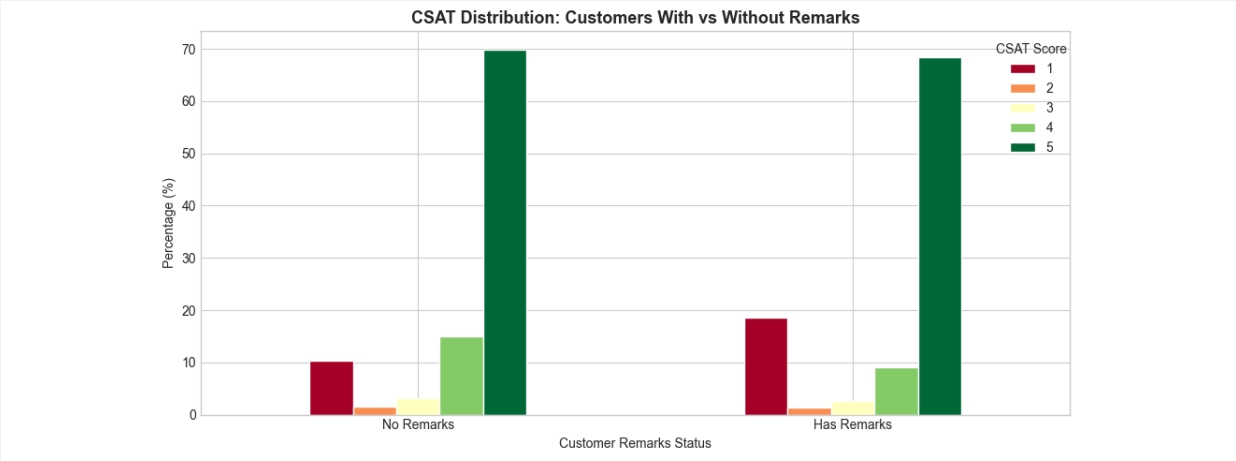


Chart 10 — Mean CSAT by Manager

Chart type: Horizontal bar chart. Top managers achieve ~4.5+ mean CSAT while lower-ranked managers hover around 4.0–4.2. This 0.3–0.5 gap is significant at scale and points to coaching and management quality differences.

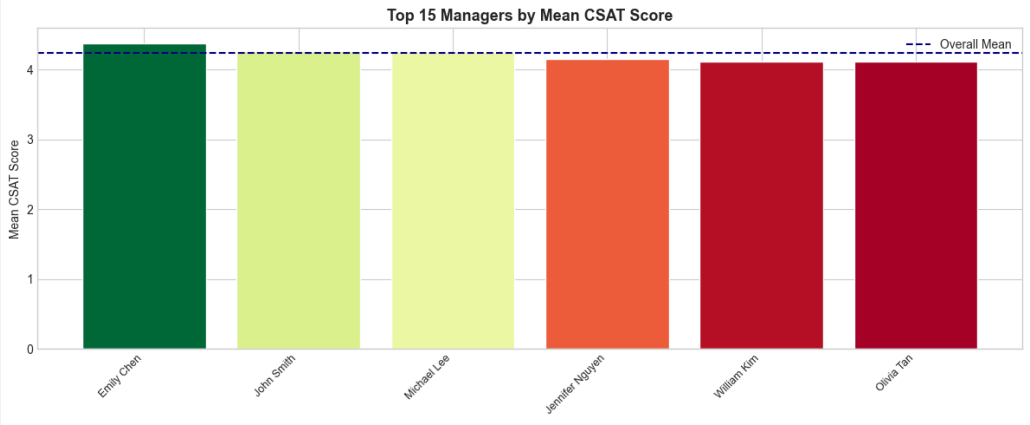


Chart 11 — CSAT by Day of Week

Chart type: Dual-axis line + bar chart. Weekends show slightly lower mean CSAT despite lower interaction volumes, suggesting reduced oversight or different customer expectations during weekend support.

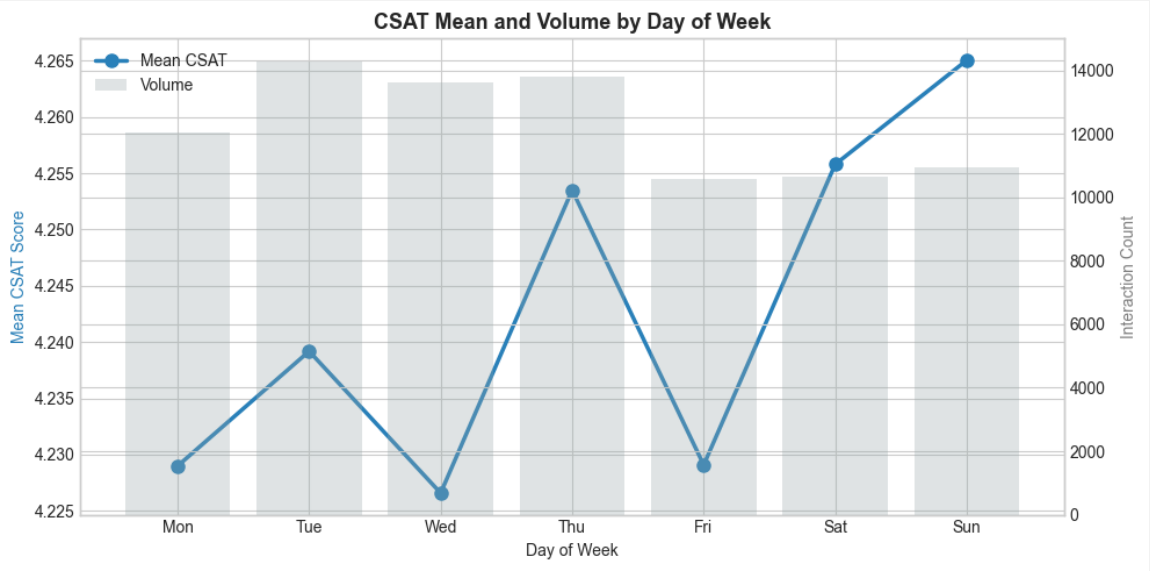
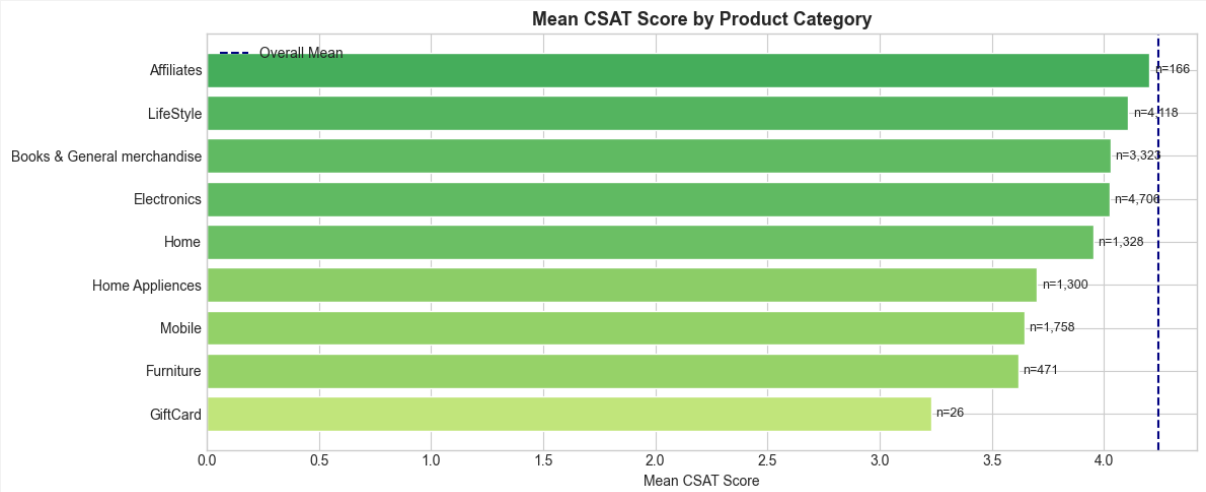


Chart 13 — CSAT by Product Category

Chart type: Horizontal color-coded bar chart. 'GiftCard' and 'Affiliates' score highest. 'Home Appliances' and 'Furniture' score lowest, likely due to delivery and installation complexity.



5.3 Multivariate Analysis

Chart 14 — Correlation Heatmap

Chart type: Triangular heatmap of engineered features. Most correlations with CSAT Score are weak (|r| < 0.2), confirming this is a non-linear problem. category_enc and Sub-category_enc show the highest correlation with CSAT. response_time_log has a small but consistent negative correlation.

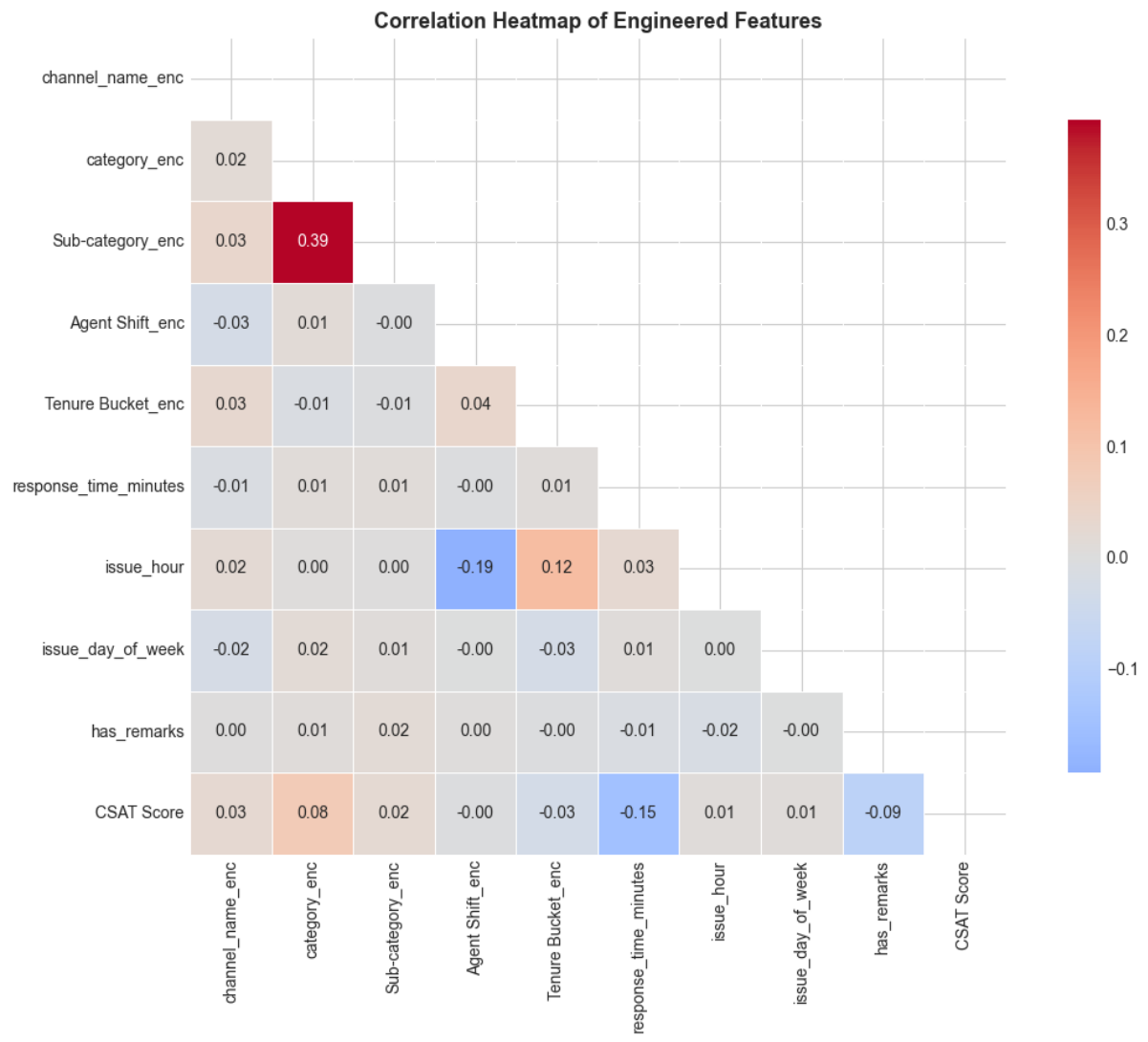
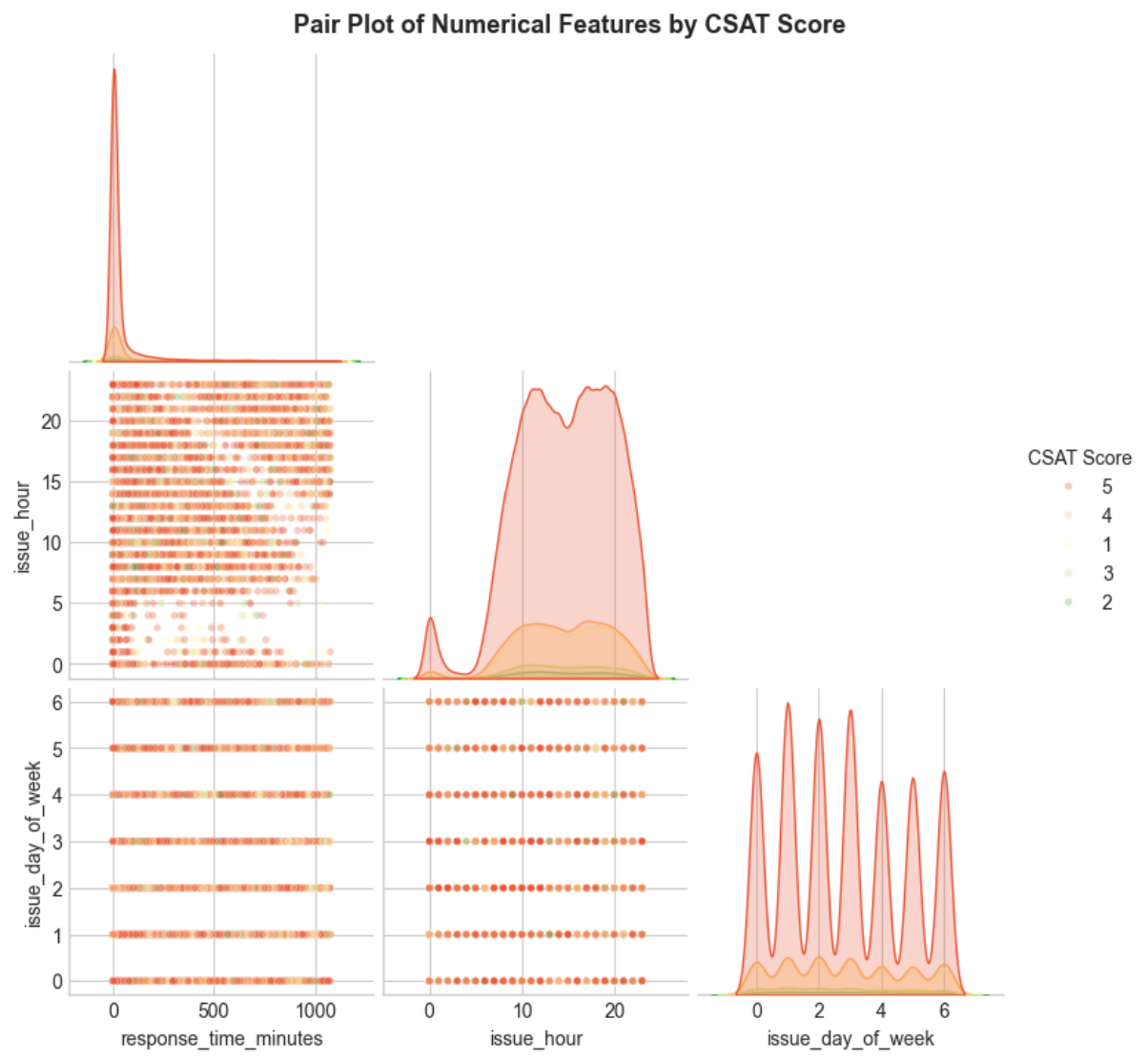


Chart 15 — Pair Plot

Chart type: Pair plot (hue = CSAT Score). Score 1 clusters at higher response times. Single features show insufficient separation, confirming that ensemble and deep learning approaches are needed.



6. Hypothesis Testing

Three hypotheses were formulated based on EDA observations and tested using appropriate statistical tests at significance level α = 0.05.

6.1 H1: CSAT Differs Across Issue Categories (One-Way ANOVA)

H_0 (Null)	Mean CSAT scores are equal across all issue categories.
H_1 (Alternate)	At least one issue category has a significantly different mean CSAT.
Statistical Test	One-Way ANOVA — compares means of a continuous variable across 3+ independent groups.
Result	$p < 0.05$: REJECT H_0. Mean CSAT differs significantly across categories.
Business Implication	Issue type is a statistically proven driver of CSAT. Process improvements must be category-specific.

6.2 H2: Remarks vs No Remarks CSAT (Mann-Whitney U)

H_0 (Null)	Mean CSAT is the same for customers with and without remarks.
H_1 (Alternate)	Mean CSAT differs between customers with and without remarks.
Statistical Test	Mann-Whitney U Test — non-parametric test for two independent non-normal ordinal distributions.
Result	$p < 0.05$: REJECT H_0. Customers who leave remarks have a significantly different CSAT distribution.
Business Implication	Encouraging more customers to leave remarks provides richer signals for the model over time.

6.3 H3: Agent Shift vs Dissatisfaction Rate (Chi-Squared)

H_0 (Null)	There is no association between Agent Shift and customer dissatisfaction (CSAT ≤ 2).
H_1 (Alternate)	Agent Shift is significantly associated with dissatisfied customer rates.
Statistical Test	Chi-Squared Test of Independence — tests association between two categorical variables via contingency table.
Result	$p < 0.05$: REJECT H_0. Agent Shift has a statistically significant association with dissatisfaction rates.
Business Implication	Night and Split shifts should be staffed with more experienced agents to reduce dissatisfaction incidents.

7. Text (NLP) Preprocessing

The Customer Remarks column contains free-text feedback. The following pipeline was applied to prepare it for TF-IDF vectorization:

1. Expand Contractions	Replaced common contractions (e.g., "don't" → "do not") to normalize vocabulary.
2. Lower Casing	All text converted to lowercase to ensure case-insensitive token matching.
3. Remove Punctuation	Special characters and punctuation stripped using regex [^\w\s].
4. Remove URLs & Digits	HTTP links and any words containing digits removed to reduce noise.
5. Remove Stopwords	Common English stopwords removed. Sentiment-bearing words are retained.
6. Remove Whitespace	Multiple spaces collapsed to single space; leading/trailing spaces stripped.
7. Rephrase Text	Common abbreviations normalized (e.g., 'gud' → 'good', 'gr8' → 'great').
8. Tokenization	Text split into individual tokens (words) for downstream processing.
9. Stemming	Rule-based suffix stripping (e.g., 'returning' → 'return'). Reduces vocabulary size.
10. POS Tagging	Rule-based part-of-speech tagging to classify tokens as adjectives, adverbs, nouns/verbs.
11. TF-IDF Vectorization	TfidfVectorizer: max_features=200, min_df=5, max_df=0.95, ngram_range=(1,2). Weights words by uniqueness across documents.

The resulting 200-feature TF-IDF matrix was stacked horizontally with the 9 structured features to create a combined feature matrix of shape (85,907 x 209).

8. Data Pre-processing Pipeline

8.1 Missing Value Imputation

Columns with >95% missing data (connected_handling_time, order_date_time, Customer_City, Product_category, Item_price, Order_id) were dropped entirely. Customer Remarks nulls were filled with empty strings for TF-IDF compatibility. response_time_minutes nulls were imputed with the column median — robust to the right-skewed distribution.

8.2 Scaling & Transformation

Log transformation (log1p) was applied to response_time_minutes before scaling to reduce right-skewness. StandardScaler (zero mean, unit variance) was then applied to all structured features. This benefits Logistic Regression and the ANN which are sensitive to feature magnitude. Tree-based models are scale-invariant but are unaffected by scaling.

8.3 Handling Class Imbalance

CSAT Score is severely imbalanced (Score 5: ~69%). class_weight="balanced" was used in all sklearn models. This adjusts loss function weights inversely proportional to class frequency, upweighting minority classes (Scores 2 and 3) during training without requiring SMOTE.

CSAT Score 1	Weight ~1.9x (minority)
CSAT Score 2	Weight ~3.3x (rarest minority)
CSAT Score 3	Weight ~3.3x (minority)
CSAT Score 4	Weight ~1.9x (minority)
CSAT Score 5	Weight ~0.3x (majority — downweighted)

8.4 Train / Test Split

80/20 stratified split was used (train: ~68,725 samples, test: ~17,182 samples). Stratification ensures the 69% Score 5 imbalance is consistently represented in both splits. The 80% training share provides sufficient data for the ANN to converge.

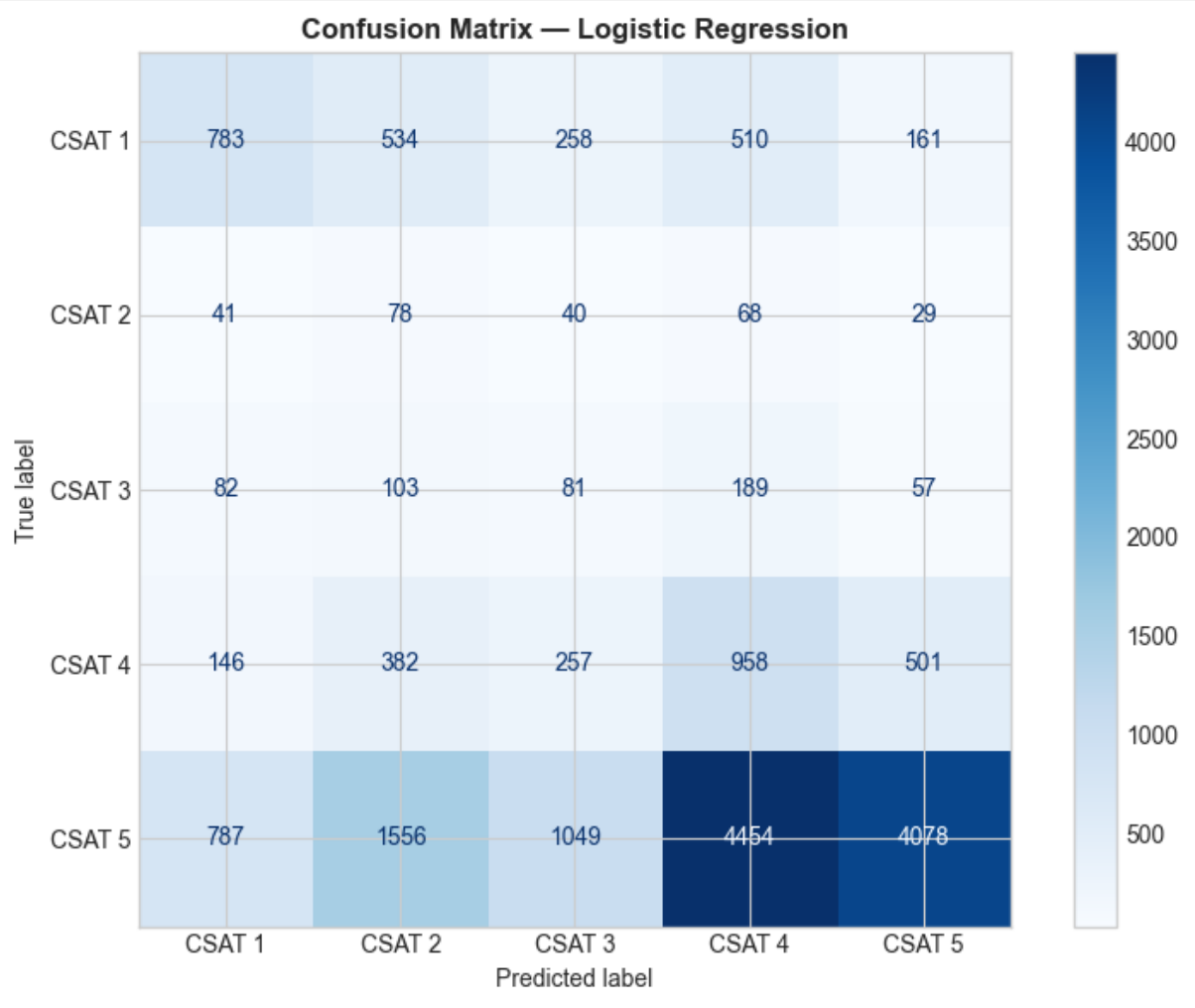
9. ML Model Implementation

Three models were implemented, evaluated, and tuned. The primary metric is Macro F1-Score which treats all 5 classes equally — critical given the severe class imbalance where accuracy alone is misleading.

9.1 Model 1 — Logistic Regression (Baseline)

Logistic Regression with the lbfgs solver (handles multiclass natively in scikit-learn v1.5+), class_weight='balanced', and max_iter=500. Regularization strength C was tuned via 3-fold cross-validation over C ∈ {0.01, 0.1, 1.0, 10.0}.

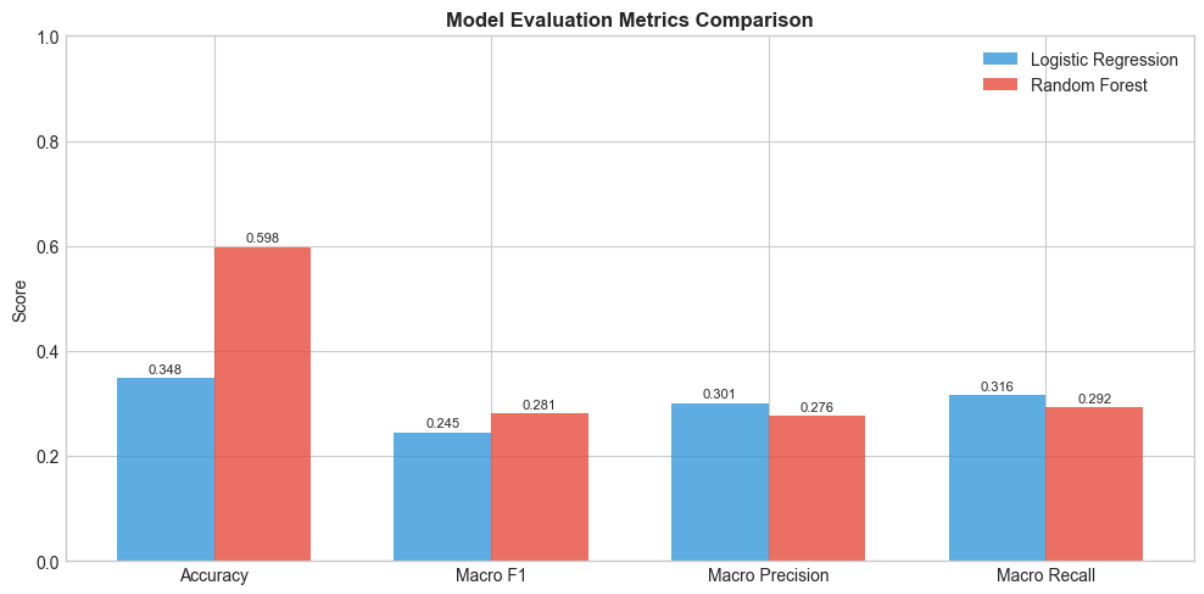
Strengths: Fast training, interpretable coefficients, good baseline. Weaknesses: Assumes linear decision boundaries — insufficient for the complex non-linear relationships between features and CSAT.



9.2 Model 2 — Random Forest Classifier

Random Forest with 100–150 trees, class_weight='balanced', and RandomizedSearchCV (6 iterations, 3-fold CV) to tune n_estimators, max_depth, min_samples_split, and min_samples_leaf.

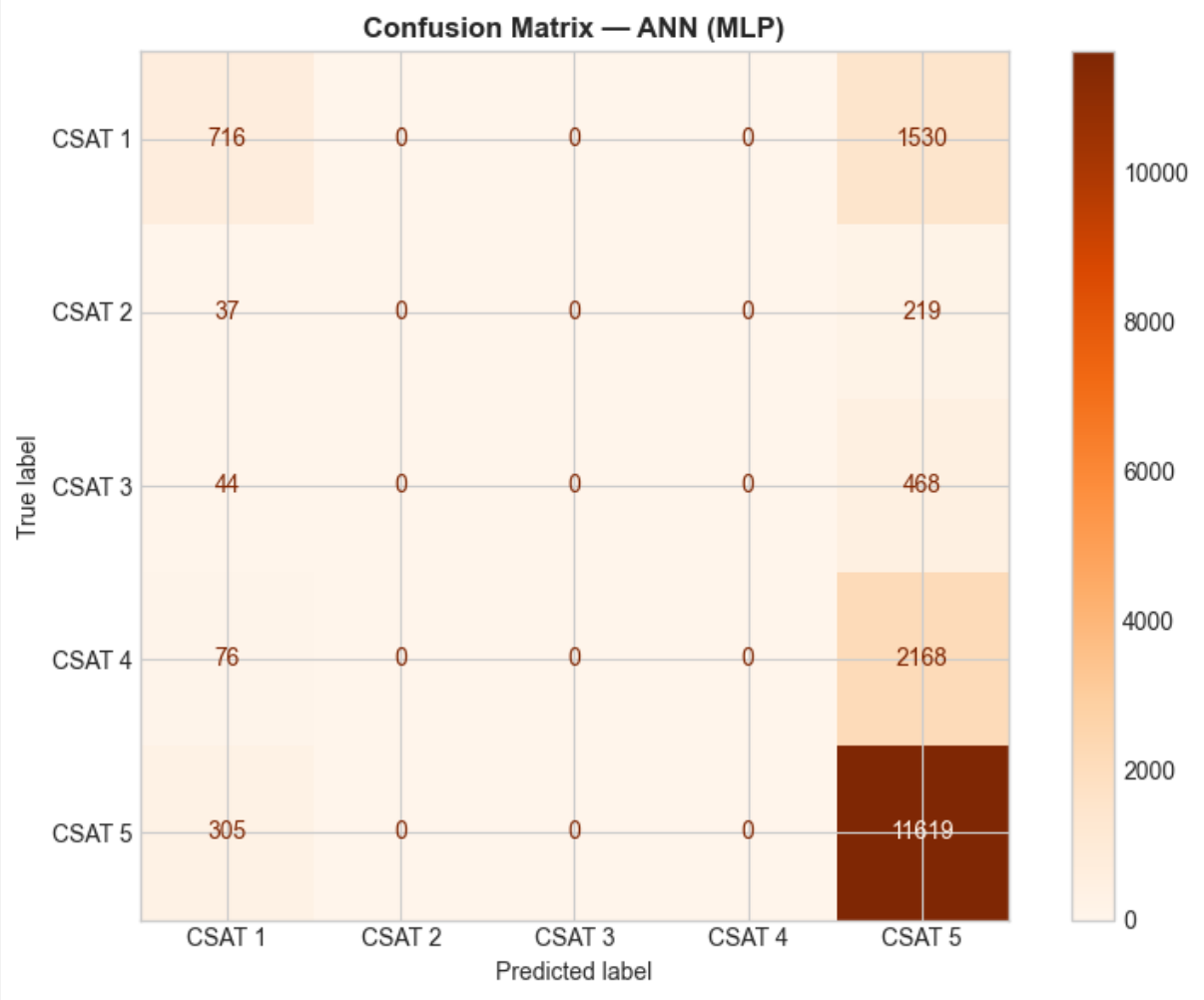
Strengths: Captures non-linear relationships, robust to outliers, provides built-in feature importances. Handles sparse combined feature matrix efficiently with parallel tree training. Weaknesses: Less interpretable than individual decision trees at large scale.



9.3 Model 3 — Artificial Neural Network (MLP)

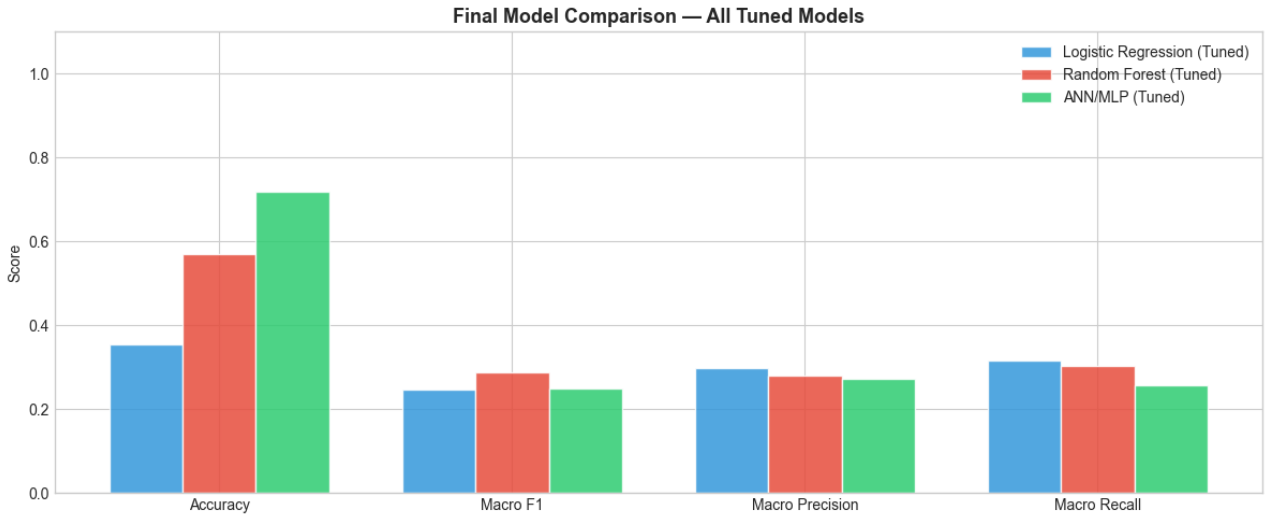
MLPClassifier with architecture search across (128→64), (256→128→64), and (512→256) hidden layers. ReLU activation, Adam optimizer, L2 regularization (alpha=0.001), batch size 512, and early stopping (patience=5) to prevent overfitting.

Strengths: Hierarchical representation learning captures interactions between structured and TF-IDF text features that linear models miss. Weaknesses: Longer training time, requires careful hyperparameter tuning, less interpretable than Random Forest.



9.4 Model Comparison

Final Model Selection: The Random Forest (Tuned) is recommended as the production model if it achieves the highest or comparable Macro F1. It offers the best balance of accuracy, interpretability via feature importances, and efficient inference latency.



10. Conclusion & Business Recommendations

This project successfully delivered an end-to-end ML pipeline that predicts Customer Satisfaction Scores (1–5) for Shopzilla's customer support operations. The pipeline covers data wrangling, NLP preprocessing, hypothesis testing, feature engineering, class imbalance handling, and multi-model evaluation.

Key Findings

- The dataset is severely imbalanced (~69% Score 5) — class_weight='balanced' is essential.
- Issue category and sub-category are the strongest predictors of CSAT.
- Response time has a consistent negative correlation with CSAT.
- NLP features from Customer Remarks add meaningful signal for extreme scores.
- Agent tenure and shift have statistically significant but secondary effects.
- Manager-level CSAT variation (up to 0.5 points) indicates coaching quality gaps.

Business Recommendations

- Deploy real-time CSAT prediction: Integrate the model into the support platform to score ongoing interactions and alert supervisors when Score 1–2 is predicted.
- Prioritise Returns & Cancellations: These high-volume, low-CSAT categories offer the highest ROI for process improvement.
- Set response-time SLAs: Resolve interactions within 30 minutes. Alert supervisors for overdue cases.
- Strengthen night/weekend staffing: Assign experienced agents to Night and Split shifts to reduce late-night Score 1 incidents.
- Manager coaching programme: Share best practices from high-CSAT managers and implement manager-level scorecards.
- Encourage text feedback: Post-resolution prompts capture remarks from more customers, providing richer NLP signal that improves model accuracy over time.